DTC P117A:00 [SKYACTIV-D 1.5]

id0102q2316900

Note
- This DTC is detected as a record indicating that the torque restriction control for piston protection has been performed.
- The torque restriction control for piston protection prevents engine burn by decreasing the fuel injection amount.

DTC P117A:00	Record of torque restriction for piston protection
DETECTION CONDITION	• The engine oil pressure is the specified value or less for a continuous **15 s**. • The vehicle is driven continuously for a certain period of time at low engine speed and high load. **Diagnostic support note** • This is an intermittent monitor (other). • The check engine light does not illuminate. • FREEZE FRAME DATA/Snapshot data is not available. • DTC is not stored in the PCM memory.
FAIL-SAFE FUNCTION	Not applicable
POSSIBLE CAUSE	• Engine oil pressure is low. • Engine oil temperature is high. • High engine speed condition is continued with the engine oil diluted. • Vehicle is driven continuously at low engine speed and high load • Engine oil pressure sensor malfunction • PCM malfunction
SYSTEM WIRING DIAGRAM	Not applicable

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
3	**INSPECT EFFECT OF VEHICLE CONDITION FOR MALFUNCTION** • Verify how the customer drives the vehicle by asking the customer the following: — Continue the high engine speed condition with the message related to a diesel particulate filter malfunction in the display. — Continue the high engine speed condition with the diesel particulate filter indicator light illuminated. — Vehicle is driven continuously at low engine speed and high load. • Are any of the conditions above applicable to the vehicle use conditions?	Yes	Explain to the customer that the vehicle is normal. (malfunction caused by driving at high engine speed with engine oil diluted) If there is a concern with customer's driving, provide the customer some pertinent advice (such as gear selection, how to use manual mode). Go to Step 6.
		No	Go to the next step.
4	**INSPECT ENGINE OIL LEAKAGE** • Start the engine. • Visually inspect the engine oil runner for engine oil leakage. • Is there any leakage?	Yes	Repair or replace the malfunctioning part according to the inspection results, then add genuine motor oil. Perform the "ENGINE OIL DATA RESET", then go to Step 6. (See ENGINE OIL DATA RESET [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
5	**INSPECT ENGINE OIL PRESSURE SENSOR** • Inspect the engine oil pressure sensor. (See ENGINE OIL PRESSURE SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the engine oil temperature sensor/engine oil pressure sensor, then go to the next step. (See ENGINE OIL TEMPERATURE SENSOR/ ENGINE OIL PRESSURE SENSOR REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
6	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Start the engine and warm it up completely. • Drive the vehicle. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
7	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.